Stroke Risk Prediction Project

PROJECT CHARTER

Team members (Group 6)
Pamela Abigail Castillo Gonzalez
Pardon Runesu
Johannes Haddad
Chantale Nzegge Mvele
Htet Wai Aung

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1. INTRODUCTION

1.1 PURPOSE OF PROJECT CHARTER

The *Stroke Risk Prediction Dashboard* project charter document the necessary information required by decision makers to approve and support the project. It defines the project's needs, scope, justification, and required resource commitment as well as the project's sponsor(s) decision to proceed or not to proceed with the project. It was created during the Initiating Phase of the project.

The intended audience of the *Stroke Risk Prediction Dashboard* project charter is the project sponsor and senior leadership.

2. PROJECT AND PRODUCT OVERVIEW

The aim of the *Stroke Risk Prediction Dashboard* project is to design and implement a user-friendly and interactive dashboard using a Stroke Prediction dataset. The dashboard would help healthcare providers, public health stakeholders, and researchers to analyze stroke risk factors, visualize trends, and predict individual stroke risk using machine learning models.

The duration of the project is 8 weeks during which team members would contribute their time and expertise as part of PROHI course work.

Most of the software used during the project is free/open-source tools (Python,GitHub, Google Docs).

2. Project and Product Overview

The project will be led by the Neuro Predict Team's Data Science and Product Development team, with support from IT and Quality Assurance. The end users of the dashboard include healthcare professionals, researchers, and public health organizations.
The deliverable is a **Stroke Risk Prediction Dashboard**, built with Streamlit, that integrates a machine learning model trained on a Kaggle stroke dataset. The dashboard will provide risk prediction, visualization of key risk factors, and an intuitive/interactive user interface for decision support.
The project will start on 2025-09-02 and is estimated to take 8 weeks to complete and the development will be conducted at Neuro Predict Team's headquarters with a distributed team collaborating remotely through GitHub and VS Code. The final dashboard will be deployed on GitHub/Streamlit Cloud for global access.

Estimated Duration: 8 weeks,
Estimated Budget: 217,325 sek

3. JUSTIFICATION

3.1 BUSINESS NEED

Stroke remains a leading cause of disability and death worldwide. Early prediction of stroke risk empowers healthcare providers to implement timely preventive measures, such as lifestyle modifications, blood pressure management, and diabetes control. The Stroke Risk Prediction Dashboard will assist healthcare providers and public health organizations in monitoring

prevalence and incidence of stroke-related risk factors in populations. The dashboard will help estimating the likelihood of stroke occurrence at the individual level, identifying changes in trends and patterns of stroke determinants and detect high-risk populations to enable targeted prevention and intervention strategies.
By providing timely, data-driven insights, the system will enhance clinical decision-making, reduce stroke-related morbidity, reduce mortality, and contribute to cost savings for healthcare systems

3.2 Public Health and Business Impact

By transforming complex patient data into actionable insights, an accessible dashboard tool supports clinicians in identifying modifiable risk factors and reducing the likelihood of severe stroke events. This in turn reduces stroke prevalence, morbidity, and mortality. It may also guide health resource allocation and preventive care strategies, reducing costs linked to long-term disability. From a public health perspective, the implementation of the dashboard can improve quality of life by preventing long-term disability, support population health monitoring and inform targeted prevention programs and decrease stroke-related morbidity and mortality through proactive interventions.

From a business perspective, the system strengthens the Neuro Predict team's position as a leader in predictive healthcare analytics. It opens opportunities for partnerships with institutions like hospitals, insurance companies, and public health agencies while demonstrating the team's ability to deliver innovative, scalable digital health solutions.

3.3 Strategic Alignment

Goal	Project Response Rank	Comments
Scale: **H** – High, **M**- Medium, **L** – Low, **N/A** – Not Applicable		
NC / Division / Branch Strategic Goals:		
Strategic Alignment (Company Level)	High	The project supports the company's core strategy of leveraging data analytics to improve health outcomes and deliver innovative healthcare tools. Aligns tightly with business missions.
The Public Health Agency (Folk Hälsomyndigheten-FHM) Digitalization-related Strategic Goals		
FHM goals (See Appendix A)	Medium	Aligns with FHM 's emphasis on increasing access to digital tools/technologies to support social participation, especially among elderly and vulnerable populations (reduce digital exclusion). Strengthen data collection, analysis, and reporting to inform public health policy — achieving transparency,

Goal	Project Response Rank	Comments
		granularity (age, region, socioeconomic etc.), timely and comparable data.
The National Board of Health and Welfare (Socialstyrelsen) Strategic & Digitalization Goals		
Relevant national board of health and welfare's strategic goals (See Appendix A)	High	The national board of health and welfare's strategic goal's (FY 2022-2026) as shown in appendix c includes goals like health care quality improvement, patient safety, strengthening innovation and the use of technology and encouraging data driven governance.
The National Board of Health and Welfare (Socialstyrelsen)'s IT Goals:		
National board of healthcare and welfare strategic IT goals (See Appendix A)	Medium	While the project uses modern tools (Streamlit, GitHub, cloud deployment), full alignment with large-scale IT infrastructure goals (e.g. enterprise interoperability, secure large-scale data exchange) with a goal: developing digital tools for specialized tools to enhance healthcare.
Karolinska university (KI)'s Digitalization goals		
Karolinska university digitalization goals (See Appendix C)	High	The goals include areas like Accountability & Transparency, IT Modernization, and "provide innovative and data-driven solutions. "The project moderately aligns with these: strong on innovation & data, less so currently on cross-agency federal coordination or regulatory compliance.

4. Scope

4.1 Objectives

The objectives of the Stroke Risk Prediction Dashboard are as follows:

- Develop a tool to identify patients at risk of stroke and support timely preventive interventions.
- Build an interactive dashboard to visualize predictions and key risk factors.
- Develop a machine learning model to predict individual stroke risk.
- Analyze the data using descriptive, diagnostic, predictive, and explainable analytics on stroke risk factors

4.2 High-Level Requirements

The following table presents the requirements that the project's product, service or result must meet for the project objectives to be satisfied:

Req. #	Requirement Description
1	The system needs to be supervised to ensure no bugs, failures, or errors occur.
2	The dashboard will provide a result of whether the patient is at risk of having a stroke or not, and some preventative measures.
3	User-friendly, ensuring the user is easily able to orient around the dashboard and access the information they give and receive.
4	Data sources and metadata provided for users as well as contact information for any questions or help needed
5	Incorporate visualizations, such as charts, that explain which features contributed most to risk assessment. These explanations should aid physicians in understanding and trusting the model's decisions.
6	The dashboard will allow the users to input data.
7	The system will comply with GDPR and all other applicable data privacy laws.

4.3 Major Deliverables

The following table presents the major deliverables that the project's product, service or result must meet for the project objectives to be satisfied. Our deliverables will follow the six main aspects of "PRINCE2" [2] to ensure our projects quality. PRINCE2 stands for "Projects in Controlled Environments" [2].

Major Deliverable	Deliverable Description
Data	Dataset with wanted data to be used as our database
Pipeline	A pipeline that shows what changes and/or corrections have been made with the chosen dataset. Handling of missing values, datatype changes, etc.
Machine learning data model	A trained data model to be used, which our dashboard will use when delivering results.
Dashboard	A simple dashboard that any user can use regardless of their technical skill level
Information and instruction manuals	Information regarding the purpose of the project and what to expect, as well as a "How to use" manuals
Presentation	A presentation to show our final project and the end results

4.4 Boundaries

Due to the time scope for this project, certain boundaries had to be made to ensure meeting our time requirements as well as quality assurance.

- Using Kaggle to find an appropriate dataset instead of using a dataset from an institution due to our time constraint.

- Excluding any other diagnosis, except Stroke
- Not enough time for adequate testing
- Not able to publish a demo for users to find bugs, errors, or missing features.
- Detailed documentation and pipeline may not be available in time
- Integration with other machine learning models, only focusing on one at this time
- Advanced functions, such as app integration and account creation, will not be made in time
- Focus more on function rather than perfect design
- For now, it is only made for one platform

5. Duration

5.1 Timeline

Based on an 8-week project duration (3) from September 1st to October 19th, 2025, the project timeline is structured as follows:

Sprint 1: Foundation & Setup (September 1-7, 2025)

- Team formation and role assignment
- Team Rules document development and submission
- Kaggle dataset download and initial review
- Stakeholder identification and engagement
- Technical environment setup and tool configuration
- Initial project documentation framework

Sprint 2: Project Charter Preparation, Enhanced Data Insights and Visualization (September 8-21, 2025)

- Incorporate feedback and lessons from Sprint 1
- Project charter development and approval
- Data quality assessment and profiling
- Exploratory data analysis and visualization
- Data cleaning and preprocessing
- Feature identification and preliminary analysis
- Build a basic, functional dashboard using Strealit that displays initial features of the dashboard

Sprint 3: User Interaction, Model Development & Optimization (September 22 - October 5, 2025)

- Incorporate feedback and lessons from Sprint 2
- Machine learning algorithm selection and implementation
- Model training, testing, and validation
- Performance evaluation and optimization
- Implement technical feedback points from the Peer Review Team and Project Board into the dashboard development process
- Model documentation and interpretation

Sprint 4: Dashboard Interactivity and User Interfacts Improvement (October 6-12, 2025)

- Incorporate feedback and lessons from Sprint 3
- User interface design and wireframing
- Model integration with the dashboard interface

- User experience testing and refinement
- Optimize interactive features to the dashboard

Sprint 5: Final Integration & Delivery (October 13-19, 2025)

- Incorporate feedback and lessons from Sprint 4
- Perform comprehensive system integration and end-to-end testing
- Documentation finalization and review
- Presentation material preparation
- Final quality assurance and bug fixes
- Project delivery and handover preparation

5.2 Executive Milestones

The table below lists the high-level Executive Milestones of the project and their estimated completion timeframe.

Executive Milestones	Estimated Completion Timeframe
Team Rules document preparation, team organized, and initial project brief approved	September 7, 2025
Project charter approved	September 21, 2025
Data acquisition and exploration completed	September 21, 2025
Machine learning models developed and validated	October 5, 2025
Dashboard implementation completed	October 12, 2025
Final project delivery and documentation	October 19, 2025
Seminar presentation and peer review	October 29, 2025

Section 6: Budget Estimate

6.1 Funding Source

The Stroke Risk Prediction Dashboard project is funded through a multi-source allocation model that leverages national healthcare investment priorities and institutional support:

Primary Funding Sources:

- **National Health Budget Allocation (65% - 141,261 SEK):** Allocated from Sweden's national health digitalization initiative, supporting the development of predictive healthcare tools that align with public health objectives
- **Karolinska Institutet Research Fund (25% - 54,331 SEK):** Institutional support for innovative health informatics projects that advance medical research capabilities

- **PROHI Course Budget (10% - 21,733 SEK):** Educational program allocation for practical learning projects that contribute to healthcare innovation

Expected Funding Sources

Funding Source	Percentage	Amount (SEK)	Purpose
National Health Budget Allocation	65%	141,261	Human Resources, Technology
Karolinska Institutet Research Fund	25%	54,331	Operational Expenses, Materials
PROHI Course Budget	10%	21,733	Miscellaneous, Contingency
TOTAL FUNDING	**100%**	**217,325**	**Complete Project Coverage**

This funding model ensures sustainable project execution while demonstrating the collaborative approach between national health policy, academic research, and educational objectives in advancing healthcare technology solutions.

6.2 Estimate

This section provides a summary of estimated spending to meet the objectives of the Stroke Risk Prediction Dashboard project as described in this project charter. This summary of spending reflects costs for the entire 8-week project lifecycle and presents probable funding requirements to assist in obtaining budgeting support [4-5].

Summary Budget Estimates

Category	Amount (SEK)	Percentage of Total
Human Resources	196,000	90.19%
Software & Technology	4,300	1.98%
Materials & Stationery	1,500	0.69%
Telecommunications	1,400	0.64%
Electricity	2,400	1.10%
Miscellaneous	1,700	0.78%
Contingency	10,025	4.61%
Total Project Budget	**217,325**	

Budget Justification: This budget reflects a comprehensive approach to developing a healthcare prediction dashboard that meets professional standards. The significant investment in human resources (90%) ensures expert-level development, while operational costs support sustainable project execution within the 8-week timeframe. The multi-source funding approach demonstrates alignment with national health digitalization priorities and institutional research objectives.

7. High-Level Alternatives Analysis

A number of high-level alternative options were considered in the planning of this project of developing a stroke prediction interactive dashboard for healthcare professionals. The summary table is prepared to compare pros and cons of each alternative.

Alternative 1: Continue the current medical practice by health professionals, and no prediction model dashboard is considered

Description: This option maintains the status quo by relying on clinical judgment to assess stroke risks in the care and treatment of patients. It requires no financial investments and does not entail any changes to the existing healthcare workflow.

Factors considered: Choosing not to take action will not enhance the current state of healthcare or assist healthcare professionals in improving the quality of care. By depending on traditional methods, patients will miss out on better risk prediction and prevention strategies, which could reduce disability and mortality rates associated with strokes, especially when using machine-learning prediction models.

To enhance healthcare for patients at risk of stroke, this option holds no value and should therefore be disregarded.

Alternative 2: Develop a simple rule-based decision support system within the Electronic Health Record (EHR) system with Guideline Definition Language (GDL)

Description: This alternative involves creating and implementing a straightforward rule-based decision support feature within the EHR system, which is based on clinical guidelines and a flowchart. This simple flowchart system uses a scoring scheme that takes into account several variables, such as blood pressure, age, BMI, blood cholesterol levels, and dietary fat condition. Based on the total score, the decision support system outputs the risk level for experiencing a stroke.

Points considered: This alternative aims to help healthcare providers, particularly new doctors or those with a heavy workload, in identifying stroke risks. However, since this decision support system relies on a simple flowchart and does not utilize advanced machine-learning prediction algorithms, it may not offer significant added value in the current healthcare landscape, especially when considering the investments and time required to develop and deploy the system. Therefore, this alternative is not recommended as a viable option for improving the care and treatment of individuals at risk for strokes.

Alternative 3: Develop a Stroke Risk Prediction decision support dashboard with a machine-learning model (recommended)

Description: This alternative aims to develop a comprehensive decision support tool that employs a machine learning model to identify patients at risk of stroke, features an interactive dashboard for visualizing predictions and key risk factors.

Point considered: This custom, highly effective, and interactive dashboard will significantly assist healthcare professionals in predicting stroke risks in high-risk patients and providing preventive measures. Although this option requires a substantial financial investment and time commitment, it is the most recommended solution to greatly enhance the quality of healthcare services for patients at high risk of stroke.

8. Assumptions, Constraints And Risks

8.1 Assumptions

This section identifies the statements believed to be true and from which a conclusion was drawn to define this project charter.

1. **Stakeholders' involvement-** Stakeholders will be available for feedback and approval of the dashboard when needed.
2. **Dataset accessibility** – The system makes use of the stroke prediction database available on kaggle and it is hoped the data will remain publicly available and downloadable throughout the duration of the project.
3. **Data representativeness** – The team assumes the dataset is representative enough to the population for building and testing predictive models, even though it may have limitations.
4. **Open-source tools availability** – All required tools and libraries (Python, GitHub) will remain freely available for use.
5. **Team engagement**- It is assumed that all team members will actively contribute to the development of the dashboard.
6. **User-friendly**- Healthcare professionals are developing the dashboard, ensuring it will be user-friendly and easy to navigate.

8.2 Constraints

This section identifies any limitations that must be taken into consideration prior to the initiation of the project.

1. **Time Constraint:** The project must be completed within 8 weeks, which limits the extent of experimentation and thoroughness of model optimization.
2. **Dataset Limitations:** The Kaggle stroke dataset contains only 11 features and is unbalanced, which may adversely affect the accuracy and generalizability of the predictive models.
3. **Scope Limitation:** The project is designed solely for educational and demonstration purposes and cannot be validated or deployed in real clinical environments.
4. **Budget Limitation:** The scope is confined to specified dashboard features; additional functionalities or enhancements cannot be accommodated within the current budget.
5. **Unexisting information about the dataset-** The dataset lacks information about the population, which may limit the generalizability of the results to all groups

8.3 Risks

Risk	Mitigation
Skill gaps among team members in GitHub, dashboard development, and machine learning may lead to project delays in delivering the final product.	Roles would be assigned based on expertise. Online resources would be used. Group members would support and help each other and organize internal learning sessions if need be and reach for the mentor to ask for help.

Risk	Mitigation
Insufficient collaboration and communication among team members may slow the project and affect team performance.	Weekly check-ins would be done on Wednesdays through Whatsapp. Teammates would be able to communicate through GitHub and the project coordinator/manager would be the overseer.
The roles and responsibilities are not clearly distributed; therefore a task might be done by multiple team members.	Roles would be defined and documented early in the project
Expansion of project scope beyond original plans may result in extended timelines, greater resource demands, and potential project delays	A clear scope of the project would be defined in this charter program, and the team would try to adhere to the predefined scope. Focus on the primary requirements of the system and maintain consistent communication with stakeholders to prevent scope creep.
Incorrect time estimation for the tasks	Tasks will be subdivided into smaller, manageable components with clear deadlines assigned. Progress evaluation would be carried out during the group's weekly meetings.
The budget is estimated incorrectly and therefore not all the main requirements are included and the product is incomplete	Prioritize open-source tools to minimize costs, request institutional support if additional costs arise.
Low acceptance of the dashboard from healthcare professionals will produce delays and budget overruns.	Ask for feedback from healthcare professionals and other key users throughout the dashboard's development. Depending on the timeline, gather additional input once the dashboard is complete to ensure it meets users' needs.
Lack of communication	Dedicated communication channel (WhatsApp set up in the beginning of the project. When coding starts, the project manager would create a repository that the group members would all have access to and be able to see progress made by groupmates
The dashboard is confusing and non-user-friendly that may lead to low adoption, requiring additional training and support to ensure effective use.	The dashboard is designed by user-centered design, and the team will provide training to any professional that will be using the dashboard, continuing feedback even after deployment will be considered to improve the product.

9. Project Organization

9.1 Roles and Responsibilities

This section outlines the key roles that support the project. Due to the small project team's nature, members hold dural responsibilities.

Name & Organization	Project Role	Project Responsibilities
Alejandro Kuratomi Hernandez Stockholm University	Project Sponsor	● Responsible for acting as the project's champion and providing direction and support to the team. ● Approves the project scope and the funding request
Pamela Abigail Castillo Gonzalez Neuro Predict Team	Project Manager/ Product Owner	● Responsible for day-to-day management of the project and accountability for managing the project within the approved constraints of scope, quality, time and cost ● Lead in formulating project scope, objectives and deliverables ● Leading in planning and organization of tasks, timelines and resources ● Promote team dynamics and ensure effective communication ● Coordinate and collaborate with stakeholders from the hiring institution/company ● Contribute to the development, testing and deployment of the Descriptive Analysis page) of the dashboard ● Accomplish high-quality, functional deliverables as per the set timeline
Pardon Runesu Neuro Predict Team	Project Administrator/ Scrum Master	● Manage project finances effectively and efficiently ● Maintain project documents and archive files as per set procedures ● Contribute to the development, testing and deployment of the Information component of the dashboard ● Accomplish high-quality, functional deliverables as per the set timeline ● Manage Sprints process and coordinate team members for Sprints events/meetings and collaborative works
Johannes Haddad Neuro Predict Team	Project Officer/ Quality Assurance Officer	● Contribute to the development, testing and deployment of the Diagnostic Analysis component of the dashboard ● Accomplish high-quality, functional deliverables as per the set timeline ● Review the functionality of all components of the product and coordinate with team members for quality improvement and assurance

Name & Organization	Project Role	Project Responsibilities
Chantale Nzegge Mvele Neuro Predict Team	Project Officer/ Communication Officer	● Contribute to the development, testing and deployment of the Preventive Analysis component of the dashboard ● Accomplish high-quality, functional deliverables as per the set timeline ● Lead external communication with stakeholders
Htet Wai Aung Neuro Predict Team	Project Officer/ Project Monitor	● Contribute to the development, testing and deployment of the Risk Prediction component of the dashboard ● Accomplish high-quality, functional deliverables as per the set timeline ● Assist the Project Manager in monitoring project progress, ensuring deliverables meet standards, identifying risks and delays, and providing feedback to team members

9.2 Stakeholders (Internal and External)

Internal stakeholders	Description (what does the stakeholder need to do?)
Physician/ Neurologist/ Nurses (end-users)	● Test the dashboard and provide feedback on its usability, user-friendliness, and potential areas for improvement. ● Validate the functionalities of the product
Project Steering Committee members	● Oversee the progress and approve requests for modifications and adjustments of the project ● Provide overall guidance for the project
Another Project Team for Peer Review	● Peer review of the Stroke Risk Prediction Dashboard project's overall scope and work ● Provide constructive feedback and inputs
Health Data Managers of the hiring Hospital/ Institution	● Coordinate with the Neuro Predict Team on the stroke dataset, progress of the project ● Coordinate between stakeholders (healthcare providers, patients) and the Neuro Predict team to understand users requirements and needs ● Analyze existing solutions to identify unmet needs and opportunities for improvement. ● Ensure the tool integrates seamlessly with Electronic Health Records (EHRs) and other healthcare systems. ● Define and implement health data privacy, GDPR and HIPAA regulations. ● Uses standards of interoperability to transfer data between the app and the EHR

External stakeholder (role)	Description (what does the stakeholder need to do?)
Public Health Agencies	● Ensure compliance with healthcare regulations and advocate for equitable access to the tool. ● Provide technical guidance and support as necessary
General public/ patients	● Express their perceptiveness, views and concerns if present so that the project team is able to consider the perspectives of patients and adapt the product as necessary
Data Security Expert	● Provide technical advice on data security measures and actions on working with the dashboard on the Streamlit platform

DEVELOPMENT APPROACH

The project will use the Agile project management approach to ensure flexibility, the ability to accommodate constant changes and strong teamwork. With its iterative, incremental nature, core values and principles [7], we believe that this approach suits our project context best for several reasons.

To start with, health risk and outcome prediction dashboard projects that utilize machine learning models, like ours, require strong collaboration with various stakeholders. The primary users of this product are healthcare professionals, and their needs and feedback are essential throughout the development process. Therefore, the project team must engage end users and stakeholders at every stage and be open to making necessary adjustments. Adopting an Agile approach will give the project team the flexibility needed to meet user requirements in each phase, ultimately resulting in a user-friendly and effective final product.

Furthermore, modern projects, especially those involving machine learning, often face numerous unknown factors and uncertainties that can arise throughout the development process [8]. We may encounter unexpected challenges during the data pipeline stages and need to adapt our original plans accordingly. Based on the performance of the best machine learning algorithm, the final prediction component of the dashboard may also require refinement. The Agile process will allow us to incorporate new insights and adjustments as we progress.

Finally, since this is a new project and we do not have extensive experience yet, the Agile approach will help us learn and adapt throughout each stage, enabling us to deliver the final product through strong teamwork and collective effort.

To implement the Agile project development approach, we will include the following stages, adapting the spiral process [9].

- Planning Phase: The process begins with the project charter document, which outlines the project scope, objectives, and the overall structure of the final product/dashboard. This charter is tailored for an Agile approach, detailing key roles such as Project Owner and Scrum Master, and will serve as the foundation for the project.
- Development Iteratively Through Sprints: The project will be divided into sprints, which are short, fixed-length iterations lasting between two and four weeks. At the end of each sprint, team members will present functional updates and increments of the dashboard, soliciting feedback and input from the team and stakeholders.

- Stabilization: During this stage, we will focus on debugging and addressing interdependence related to each team member's outputs.
- Finalization and Release: The full functionality will be tested through various methods, and the dashboard will be finalized as the end product to be delivered to the healthcare institution.

10. PROJECT CHARTER APPROVAL

The undersigned acknowledge they have reviewed the project charter and authorized and fund the Stroke Risk Prediction Dashboard project. Changes to this project charter will be coordinated with and approved by the undersigned or their designated representatives.

Signature: ______________________ Date: ____________
Print Name: ______________________
Title: ______________________
Role: ______________________

Signature: ______________________ Date: ____________
Print Name: ______________________
Title: ______________________
Role: ______________________

Signature: ______________________ Date: ____________
Print Name: ______________________
Title: ______________________
Role: ______________________

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APPENDIX A: DOCUMENTATION

Annex 1: Folk Hälsomyndighens Digitalization-related Strategic Goals

1. **Strengthen health data systems**
    - Develop and maintain modern digital systems for disease surveillance, health monitoring, and population statistics.
    - Ensure data quality, timeliness, and interoperability across healthcare and public health.
2. **Enable secure data sharing & interoperability**
    - Contribute to national digital infrastructures (together with E-hälsomyndigheten, Socialstyrelsen, SKR).
    - Promote common standards, APIs, and frameworks that make it easier to exchange health data between regions, healthcare providers, and research.
3. **Improve preparedness and crisis response**
    - Use digital tools for real-time monitoring of infectious diseases and public health threats.
    - Ensure data systems support early warning and decision-making during outbreaks or crises.
4. **Support research, innovation & AI use**
    - Provide access to anonymized health data for research and innovation.
    - Explore digital methods, AI, and advanced analytics to improve public health insights.
5. **Promote transparency & citizen access**
    - Strengthen individuals' access to their own health information.
    - Ensure ethical and secure handling of sensitive health data, aligned with GDPR and upcoming EU Health Data Space.

Socialstyrelsen's Strategic Goals

1. **Best in the world at using digitalization for health and welfare (by 2025)**
    - Citizens, professionals, decision-makers, and researchers should all benefit from digital services.
2. **Strengthen the individual's empowerment and access**
    - People should have secure digital access to their health information and services.
    - Digital tools should enable self-care and participation in their own treatment.
3. **Improve quality, safety, and efficiency in health and social care**

  - Digital solutions should reduce administrative burden, support clinical decision-making, and minimize risks.

4. **Enable secure and efficient data sharing**

  - Build a **national digital infrastructure** for sharing health data across regions and sectors.
  - Promote standards and interoperability to ensure systems work together.

5. **Support innovation, research, and new technologies (AI, advanced analytics)**

  - Enable access to anonymized health data for research.
  - Drive innovation in healthtech and digital care models.

6. **Ensure trust, security, and legal clarity**

  - Protect personal data and strengthen cybersecurity.
  - Align with GDPR and EU's **European Health Data Space (EHDS)** initiative.Socialstyrelsen — Strategic & Digitalization Goals:(Socialstyrelsen)

Socialstyrelsen (The National Board of Health and Welfare) also has a set of goals / roles in the digitalization of health, care, and social services. Their aims are complementary to E-hälsomyndigheten. Key goals include:

1. Nationell Informationsstruktur (NI)

  - Maintain, update, and continue developing the *nationell informationsstruktur* (national information structure) to ensure semantic interoperability **—** so that data / information means the same things across systems.
  - Ensure NI adapts to new legal/regulatory developments (e.g. EHDS, laws on unified care documentation).

2. Digital Solutions & Welfare Technology

  - Promote and follow up on digital tools and services (e.g. remote monitoring, apps, reminder tools, digital assistance) in both regional health care and municipal social care.
  - Reduce digital exclusion: ensure that digital services are accessible to various population groups.

3. Quality, Safety, & Accessibility

  - Use digital solutions to increase patient / user participation, improve quality of care, and strengthen patient safety.

  - Emphasis on regulatory oversight, ethical & juridical support around digital tools. Socialstyrelsen+1

4. Efficiency and Capacity Management

  - Promote frameworks for production & capacity management in health care, with structured planning across strategic, tactical, operational levels to improve system responsiveness. Digital tools help in monitoring, forecasting, and balancing capacity.

5. Strategy for Welfare Technology (välfärdsteknik)

    - Develop and implement a strategy for welfare tech to support older people and people with special needs. Ensure welfare tech is evidence-based, safe, effective, inclusive.

6. Support for Providers & Monitoring of Use

    - Socialstyrelsen monitors how digital tools are used by regions and municipalities; identifies gaps, best practices.
    - Provides guidance / best practice under "God och När Vård" etc. (Good and close care).

Annex 2: Karolinska's Digitalization / IT Goals & Strategy

1. Karolinska University Hospital – New RDE Strategy
   As of March 2025, Karolinska University Hospital adopted a new *Research, Development, and Education (RDE) Strategy*. Some of the strategic goals in this include:

    - Improving access to health data. Karolinska Institutet Nyheter
    - Creating a central function for healthcare data sources and a common platform for data release. Karolinska Institutet Nyheter
    - Doubling external research funding over the next 10 years. Karolinska Institutet Nyheter
    - Enhancing early career development in research. Karolinska Institutet Nyheter
      So part of their digitalization goal is clearly to make data more accessible, usable, and to integrate it better in research together with care. Karolinska Institutet Nyheter

2. Health Data Integration / Interoperability
   Karolinska University Hospital is implementing *Tietoevry Care's Lifecare Open Platform*, which is aimed at integrating clinical data from different systems. This involves standardizing data formats (using openEHR), so different systems can share data more seamlessly. Tietoevry

    - The aim is to reduce fragmentation of data, make data available for clinical decisions without having to switch between multiple systems. Tietoevry

    - Also to use historical patient data and new data in a shared repository to support care, diagnostics, and research. Tietoevry

3. Digitalization in Pathology & Cancer Diagnostics
   Karolinska has a goal/project to digitize pathology work (tumour diagnostics, molecular pathology). For example, replacing microscope examinations with digital screens, sharing slide images among labs, and streamlining workflows among multiple pathology labs. Cancercentrum

4. Digital Archiving & Records
   Karolinska Institutet has projects to enhance digital archiving, for example adopting RODA (a digital archiving solution) to manage educational materials, research data, and other important records. Digital Strategy
   Also, Karolinska has moved to a digital diary system (digitial diarieföring) replacing paper-based archiving / record-keeping for many administrative documents, to improve efficiency, accessibility, and legal compliance. Karolinska Institutet Nyheter

5. Infrastructure & IT Governance
   Karolinska's *IT Office* has a mission that includes several digitalization goals:

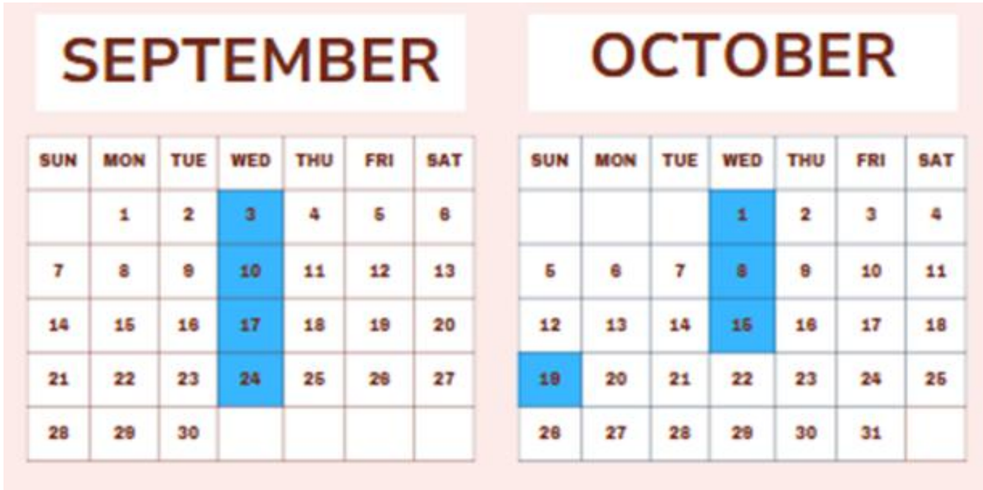
   - To maintain IT security, modern standard services, effective common IT systems. Karolinska Institutet

   - To drive digitalization of support services. Karolinska Institutet

   - To promote innovation and competitiveness in research & education through digital services. Karolinska Institutet

   - To develop KI's IT strategy. Karolinska Institutet

Annex 3: scheduling table



SEPTEMBER

SUN	MON	TUE	WED	THU	FRI	SAT
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

OCTOBER

SUN	MON	TUE	WED	THU	FRI	SAT
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Annex 4- Work Breakdown Structure (WBS) [10]

Work Breakdown Structure With Gantt Chart

Project Title	Stroke Risk Prediction Project
Project Manager	Pamela Abigail Castillo Gonzalez
Company Name	Neuro Predict Team
Date	19-Sep-2025

Sprint	Week	Days
Sprint 1	1-Sep to 7-Sep	M T W R F
Sprint 2	8-Sep to 14-Sep	M T W R F
	15-Sep to 21 Sep	M T W R F
Sprint 3	22-Sep to 28 Sep	M T W R F
	29-Sep to 5-Oct	M T W R F
Sprint 4	6-Oct to 12-Oct	M T W R F
Sprint 5	13-Oct to 19 Oct	M T W R F

WBS Number/ Sprint	Task Title
1	Project Conception, Foundation and Set-up
1.1	Team Rules Docuement
1.2	Define Roles and Responsbilities
1.3	Dataset exploration and testing
1.4	Set up technical enviroment (Github)
1.5	Sprint and Feedback Records
2	Project Charter and Data Pipelines
2.1	Project Charter Document Preparation
2.2	Data quality assessment and profiling
2.3	Exploratory data analysis and visualization
2.4	Data cleaning and preprocessing
2.5	Feature identification and preliminary analysis
2.6	Build initial feastures in the dashboard
2.7	Commuication Plan
2.8	Risk Management Plan
3	User Interaction, Model Development & Optimization
3.1	Machine learning algorithm selection and implementation
3.2	Model training, testing, and validation
3.3	Performance evaluation and optimization
3.4	Implement technical feedback points from the Peer Review Team and Project Board into the dashboard development process
3.5	Model documentation and interpretation
4	Project Performance / Monitoring
4.1	Incorporate feedback and lessons from Sprint 3
4.2	User interface design and wireframing
4.3	Model integration with the dashboard interface
4.4	User experience testing and refinement
4.5	Optimize interactive features to the dashboard
5	Project Performance / Monitoring
5.1	Incorporate feedback and lessons from Sprint 4
5.2	Perform comprehensive system integration and end-to-end testing
5.3	Documentation finalization and review
5.4	Presentation material preparation
5.5	Final quality assurance and bug fixes
5.6	Project delivery and handover preparation

Annex 5 - Detailed Budget Breakdown

Budget Category	Units	Rate (SEK)	Amount (SEK)
1. OPERATIONAL EXPENSES			
Electricity Costs	480 kWh	3	1,200
Computer power consumption (5 computers × 320 hours)	1,600 hours	1	1,200
OPERATIONAL EXPENSES Subtotal			***2,400***
2. TELECOMMUNICATIONS			
Internet Connectivity	2 months	500	1,000
Mobile Communication	2 months	200	400
Telecommunications Subtotal			***1,400***
3. STATIONERY & MATERIALS			
A4 Paper & Printing	2,000 sheets	0	200
Pens, Markers, Notebooks	Various	-	300
Poster Printing (A0 size)	2 posters	400	800
Documentation Binding	10 copies	20	200
Materials Subtotal			***1,500***
4. HUMAN RESOURCES			
Project Manager	70 hours	600	42,000
Project Officer/Scrum Master	70 hours	550	38,500
Project Officer/ QA Officer	70 hours	550	38,500
Project Officer/ Communication Officer	70 hours	550	38,500
Project Officer/ Project Monitor	70 hours	550	38,500
Human Resources Subtotal	*340 hours*		***196,000***
5. SOFTWARE & TECHNOLOGY			
Development Environment Licenses	5 licenses	400	2,000
Cloud Computing Resources	2 months	750	1,500
Collaboration Tools (Premium)	2 months	400	800
Software Subtotal			***4,300***
6. MISCELLANEOUS EXPENSES			
Meeting Room Bookings	10 sessions	50	500
Team Refreshments	8 weeks	100	800

Transportation	Various	-	400
Miscellaneous Subtotal			***1,700***
PROJECT SUBTOTAL			**207,300**
Contingency Fund (5%)			**10,025**
TOTAL PROJECT BUDGET			**217,325**

Annex 6 - Risk Management Tool [11]

Risk Management Plan											
Project Name: Stroke Risk Prediction Project							Project Manager: Pamela Abigail Castillo Gonzalez				
				Probability:	Impact:						
				5 > 75%	5 = Critical	Priority Score = Probability Score x Impact Score					
				4 > 50%	4 = Severe	High Priority Score (Red) - Response Plan required					
				3 > 25%	3 = Major	Medium Priority Score (Yellow) - Response Plan as needed					
				2 > 10%	2 = Moderate	Low Priority Score (Green) - Response Plan optional					
				1 > 0%	1 = Minor						
	Who	When	What				Who	What	When	How	Other
#	Who Identified	Date Identified	Risk Description	Probability	Impact	Priority Score	Owner	Response Type	Due Date	Response Plan	Comments/Status
1	Chantale	5-Sep-20205	Skill gaps among team members in GitHub, dashboard development, and machine learning	3	2	6	Pamela	Reduce			
2	Chantale	5-Sep-20205	Insufficient collaboration and communication among team members may slow the project and affect team performance.	2	2	4	Pamela	Reduce			
3	Chantale	5-Sep-20205	The roles and responsibilities are not clearly distributed, therefore a task might be done by multiple team members.	2	2	4	Pamela	Reduce			
4	Chantale	18-Sep-20205	Expansion of project scope beyond original plans may result in extended timelines, greater resource demands, and potential project delays	2	2	4	Pamela	Reduce			
5	Chantale	18-Sep-20205	Incorrect time estimation for the tasks	2	2	4	Pamela	Reduce			
6	Chantale	18-Sep-20205	The budget is estimated incorrectly and therefore not all the main requirements are included and the product is incomplete	2	4	8	Pamela	Reduce			
7	Chantale	18-Sep-20205	Low acceptance of the dashboard from healthcare professionals will produce delays and budget overruns.	5	5	25	Pamela	Aviod	6-Oct-2025	Gather feedback from healthcare professionals and key users during development and after completion of the dashboard to ensure it meets their needs.	
8	Chantale	18-Sep-20205	The dashboard is confusing and non-user-friendly that may lead to low adoption, requiring additional training and support to ensure effective use.	4	5	20	Pamela	Aviod	6-Oct-2025	- Implement user-centered design for the dashboard. - Provide training for all users. - Gather ongoing feedback post-deployment for continuous improvement.	